

Linear Algebra

Chapter 1 - Sets, Mappings and Complex Numbers

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Overview

1 Sets and set operations

2 Mappings

3 Algebraic Structures

4 Complex Numbers



Overview

1 Sets and set operations

2 Mappings

3 Algebraic Structures

4 Complex Numbers



- 1) A set is a well-defined collection of distinct objects. The objects that make up a set (also known as the set's elements or members) can be anything: numbers, people, letters of the alphabet, other sets, and so on.
- 2) Sets are conventionally denoted with capital letters.

Example

- A set of students in a university
- A set of Vietnamese football clubs
- The set of real numbers, denoted by $\mathbb{R}$



Basic notations

- 1) Let E be a set. If a is an element of the set E , then we denote by $a \in E$. Otherwise, we denote by $a \notin E$.
- 2) The empty set is a set containing no objects. It is written as a pair of curly braces with nothing inside $\{\}$ or by using the symbol $\emptyset$.
- 3) There are two ways of describing or specifying the members of a set.
 - i) The first way is by intensional definition, using a rule or semantic description:
 - ① A is the set whose members are the first four positive integers.
 - ② B is the set of colors of the French flag.
 - ii) The second way is by extension, that is, listing each member of the set.
 - ① $A = \{4, 2, 1, 3\}$
 - ② $B = \{\text{blue}, \text{white}, \text{red}\}$.

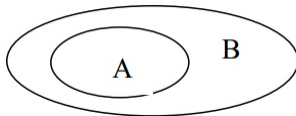


Subset, two equal sets

1) Subset

By logical expression: $A \subset B \Leftrightarrow (\forall x, (x \in A) \rightarrow (x \in B))$

By Venn diagram:



Note that $\emptyset \subset A, \forall A$

2) Two equal sets

By logical expression: $A = B \Leftrightarrow A \subset B \wedge B \subset A$

Example: $A = \{1, 2, 3, 4\}, B = \{1, 2, 3, 4, 5, 6\}, C = \{x \in \mathbb{N} | 0 < x < 5\}$

Hence, $A \subset B$ and $A = C$

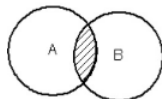


Operations on sets

Let A, B be two sets.

- ① Intersection $A \cap B = \{x | x \in A \text{ and } x \in B\}$

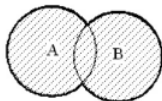
$$\begin{cases} x \in A \cap B \Leftrightarrow x \in A \text{ and } x \in B \\ x \notin A \cap B \Leftrightarrow x \notin A \text{ or } x \notin B \end{cases}$$



By Venn diagram:

- ② Union $A \cup B = \{x | x \in A \text{ or } x \in B\}$

$$\begin{cases} x \in A \cup B \Leftrightarrow x \in A \text{ or } x \in B \\ x \notin A \cup B \Leftrightarrow x \notin A \text{ and } x \notin B \end{cases}$$



By Venn diagram:

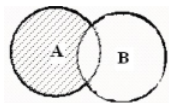


Operations on sets

Let A, B be two sets.

- 1) The difference of two sets $A \setminus B = \{x | x \in A \text{ and } x \notin B\}$

$$\begin{cases} x \in A \setminus B \Leftrightarrow x \in A \text{ and } x \notin B \\ x \notin A \setminus B \Leftrightarrow x \notin A \text{ or } x \in B \end{cases}$$



By Venn diagram:

- 2) The symmetric difference of two sets $A \triangle B = A \setminus B \text{ or } B \setminus A$
- 3) Complement $\bar{A}$

If $A \subset X$, then $\bar{A} = X \setminus A$ and is called the absolute complement of A



Exercise

Let

$$A = \{x \in \mathbb{R} \mid x^2 - 4x + 3 \leq 0\}, B = \{x \in \mathbb{R} \mid |x - 1| \leq 1\}$$

and

$$C = \{x \in \mathbb{R} \mid x^2 - 5x + 6 \leq 0\}$$

Determine

- a) $(A \cup B) \cap C$
- b) $(A \cap B) \cup C$
- c) $(A \setminus B) \setminus C$
- d) $(A \cap B) \setminus C$
- e) $(A \cup B) \setminus C$



Properties of sets

1) Commutative law:

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

2) Associative law:

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

3) Distributive law:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$



4) Property of the complement

If $A, B \subset X$, then $A \setminus B = A \cap \overline{B}$

5) De Moorgan's law

$$\overline{A \cap B} = \overline{A} \cup \overline{B}, \quad \overline{\cap A_i} = \cup \overline{A_i}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{\cup A_i} = \cap \overline{A_i}$$



The Cartesian product

Definition

- ① Let A, B be two sets. The Cartesian product of A and B , denoted by $A \times B$, is given by

$$A \times B = \{(x, y) | (x \in A) \text{ and } (y \in B)\}$$

- ② Let $A_1, A_2, \dots, A_n$ be sets. The Cartesian product of $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$ is given by

$$A_1 \times A_2 \times \dots \times A_n = \{(x_1, x_2, \dots, x_n) | x_i \in A_i, \text{ where } i = \overline{1, n}\}$$

Note that if $A_1 = A_2 = \dots = A_n = A$, then

$$A_1 \times A_2 \times \dots \times A_n = A^n$$



The Cartesian product

Definition

- ① If $(a, b), (c, d)$ are two elements of $A \times B$, then

$$(a, b) = (c, d) \Leftrightarrow \begin{cases} a = c \\ b = d \end{cases}$$

- ② If $(a_1, \dots, a_n), (b_1, \dots, b_n)$ are two elements of $A_1 \times A_2 \times \dots \times A_n$,

$$\text{then } (a_1, \dots, a_n) = (b_1, \dots, b_n) \Leftrightarrow \begin{cases} a_1 = b_1 \\ \dots \\ a_n = b_n \end{cases}$$

Examples

Let $A = \{1, 2\}, B = \{a, b, c\}$.

- a) Determine $A \times B; B \times A; A^2; B^2$.
- b) Which set does the element $(a, 1, b)$ belong to?

The set equality

Four possible methods to prove two sets are equal

- 1 Double inclusion
- 2 By using the properties
- 3 Membership table
- 4 By using Venn diagram

Examples

Prove that

- a) $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$
- b) $(A \setminus B) \setminus C = A \setminus (B \cup C)$



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Definition

Definition

Let X, Y be two nonempty sets. A map f from domain X to domain Y is an assignment of every element $x \in X$ to a **unique element** $y \in Y$, denoted as:

$$f : X \rightarrow Y$$

$$x \mapsto y = f(x)$$

Examples

- ① $f : \mathbb{R} \mapsto \mathbb{R}; f(x) = \sin x, \forall x \in \mathbb{R}$
- ② $f : X \mapsto X; f(x) = x \forall x \in X$. This is called the identity map on the set X , denoted by I_X



Definition

Let $f : X \mapsto Y$ be a map and $A \subset X$, $B \subset Y$.

1) The image

$$f(A) = \{y \in Y \mid y = f(x) \text{ and } x \in A\} = \{f(x) \mid x \in A\}$$

2) The preimage

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$$

$$x \in f^{-1}(B) \Leftrightarrow f(x) \in B$$



Examples

Ex1. Given a map $f : \mathbb{R} \setminus \{-1\} \mapsto \mathbb{R}$, $f(x) = \frac{2x+3}{x+1}$. Compute

a) $f((-1; 0])$

b) $f^{-1}([4; 7])$

Ex2. Let $f : X \mapsto Y$.

a) $f(A \cup B) = f(A) \cup f(B)$; $A, B \subset X$

b) $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$; $A, B \subset Y$.

Ex3. Let $f : \mathbb{R}^2 \mapsto \mathbb{R}^2$, $f(x, y) = (2x, 2y)$ and $A = \{(x, y) \in \mathbb{R}^2 \mid (x-4)^2 + y^2 = 4\}$. Find $f(A)$, $f^{-1}(A)$.



Injective, Surjective, Bijective maps

Let $f : X \mapsto Y$ be a map

1) The map f is called injective

- i) $\forall x_1, x_2 \in X$, if $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$, or
- ii) $\forall x_1, x_2 \in X$, if $f(x_1) = f(x_2)$ then $x_1 = x_2$, or
- iii) $\forall y \in Y$, the equation $y = f(x)$ has **at most one solution**

2) The map f is called surjective

- i) $Y = f(X)$, or
- ii) $\forall y \in Y$, $\exists x \in X$, $y = f(x)$, or
- iii) $\forall y \in Y$, the equation $y = f(x)$ has **at least one solution**

3) The map f is called bijective

- i) if it is both injective and surjective, or
- ii) the equation $y = f(x)$ has **a unique solution**



Injective, Surjective, Bijective maps

Examples

Which of the following maps are injective, surjective, or bijective?

- 1 $f : \mathbb{R} \mapsto \mathbb{R}, f(x) = 2x - 3$
- 2 $f : \mathbb{R} \mapsto \mathbb{R}, f(x) = 2|x| - 3$
- 3 $f : \mathbb{R} \setminus \{0\} \mapsto \mathbb{R}, f(x) = \frac{1}{x}$
- 4 $f : (1, +\infty) \mapsto (-1, +\infty), f(x) = x^2 - 2x$



Composition of maps

Definition

Let $f : X \mapsto Y$, $g : Y \mapsto Z$ be two maps. The map $h : X \mapsto Z$ which is defined by $h(x) = g(f(x))$, $\forall x \in X$ is called the composition of g and f . We denote by $h = g \circ f$.

Examples

Given two following maps $f : \mathbb{R} \mapsto \mathbb{R}$, $x \mapsto f(x) = x^2$ and $g : \mathbb{R} \mapsto \mathbb{R}$, $x \mapsto g(x) = 2x - 1$. Determine $f \circ g$ and $g \circ f$?

Remark

In general, $f \circ g \neq g \circ f$.



Inverse map

Definition

Let $f : X \mapsto Y$ be a bijective map. The map $f^{-1} : Y \mapsto X$ such that $f^{-1}(y) = x \Leftrightarrow f(x) = y$ is called the inverse map of f .

Properties

$$\textcircled{1} \quad h \circ (g \circ f) = (h \circ g) \circ f$$

$$\textcircled{2} \quad f \circ \text{Id} = \text{Id} \circ f = f$$

$$\textcircled{3} \quad (f^{-1})^{-1} = f$$

$$\textcircled{1} \quad f \circ f^{-1} = f^{-1} \circ f = \text{Id}$$

$$\textcircled{2} \quad (g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Injectivity, Surjectivity, Bijectivity of the composition of maps

- If f and g are injective, then $g \circ f$ is injective;
- If f and g are surjective, then $g \circ f$ is surjective;
- If f and g are bijective, then $g \circ f$ is bijective.

Exercises

Let $f : X \mapsto Y$ and $g : Y \mapsto Z$ be two maps. Prove that

- 1 if f is surjective and $g \circ f$ is injective, then g is injective.
- 2 if g is injective and $g \circ f$ is surjective, then f is surjective



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Binary operations

Definition

Let G be a set. A binary operation is a map of sets:

$$* : G \times G \rightarrow G$$

For ease of notation we write $*(a, b) = a * b \forall a, b \in G$

When we are talking about a set G , together with a fixed binary operation $*$, we often write $(G, *)$

Example

Let $B = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. Determine whether the operations of addition and multiplication are binary operations on B .



Binary operations

Properties of binary operations

Let $*$ be a binary operation on G

- 1) $*$ is said to be commutative if $x * y = y * x \forall x, y \in G$.
- 2) $*$ is said to be associative if $(x * y) * z = x * (y * z) \forall x, y \in G$.

Examples

Determine whether the following binary operation $*$ is commutative, associative?

- a) $x * y = xy - 2 \forall x, y \in \mathbb{R}$
- b) $x * y = x^y \forall x, y \in \mathbb{R}^+$
- c) $(x_1, x_2) * (y_1, y_2) = (x_1 + y_1, x_2 + y_2) \forall (x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$



Definition

A group is a set G , together with a binary operation $*$, such that the following holds:

- 1) (Associativity): $(a * b) * c = a * (b * c) \forall a, b, c \in G$
- 2) (Existence of identity): $\exists e \in G$ such that $a * e = e * a = a \forall a \in G$
- 3) (Existence of inverses): Given $a \in G$, $\exists b \in G$ such that $a * b = b * a = e$. The inverse element is denoted by a'

If $(G, *)$ satisfies only (1), then G is called a semigroup.

Definition

A group $(G, *)$ is called Abelian if it also satisfies $x * y = y * x \forall x, y \in G$



Properties of group

Properties

- 1) the identity element e is unique
- 2) the inverse element a' of a is unique

Examples

Which of the following are groups?

- 1) $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{N}, +)$
- 2) $(\mathbb{A} = \{\pm 1\}, .)$, $(\mathbb{Q}^*, .)$, $(\mathbb{R}^*, .)$



Definition

A ring is a set R with two binary operations $*$ and $\circ$ such that:

- 1) $(R, *)$ is an Abelian group
- 2) The binary operation $\circ$ is associative $\forall x, y, z \in R$:

$$(x \circ y) \circ z = x \circ (y \circ z)$$

- 3) $*$ and $\circ$ are related by the distributive law $\forall x, y, z \in R$:

$$(x * y) \circ z = x \circ z * y \circ z$$

$$x \circ (y * z) = x \circ y * x \circ z$$

- i) The ring is called commutative if the binary operation $\circ$ is commutative.
- ii) If ring R has the identity for the binary operation $\circ$, then R is called a ring with identity.

Examples

- a) $(\mathbb{Z}, +, \cdot), (\mathbb{Q}, +, \cdot)$ are commutative rings with identity.
- b) $(\mathbb{N}, +, \cdot)$ is not a ring.

Exercises

Determine whether the following sets with addition and multiplication are rings ?

- i) $X = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$
- ii) $X = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$



Definition

Let $(F, *, \circ)$ be a ring and e be the identity on $*$. We say that F is a field if

- i) F is a commutative ring with identity 1 on $\circ$, where $1 \neq e$
- ii) $\forall a \in F, a \neq e$, there always exist the inverse element x^{-1} such that $x \circ x^{-1} = 1$

Examples

- a) $(\mathbb{Q}, +, \cdot)$ is a field.
- b) $(\mathbb{Z}, +, \cdot)$ is not a field.



Exercises

Determine whether the following sets with addition and multiplication are fields?

i) $X = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$

ii) $X = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$



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Introduction

Let $\mathbb{C} = \mathbb{R} \times \mathbb{R} = \{(a; b) | a, b \in \mathbb{R}\}$ be a set such that:

$$\text{i) } (a; b) = (c; d) \Leftrightarrow \begin{cases} a = c \\ b = d \end{cases}$$

ii) Two binary operations $+$ and $\cdot$ on $\mathbb{C}$ are defined as follows:

$$\text{a) } (a; b) + (c; d) = (a + c; b + d)$$

$$\text{b) } (a; b) \cdot (c; d) = (ac - bd; ad + bc)$$



Field of the complex numbers

Proposition

The set $\mathbb{C}$ with two above binary operations $+$ and $\cdot$ is a field. We call it the field of complex numbers.

Remarks

- i) The additive identity is $(0; 0)$.
- ii) The multiplicative identity is $(1; 0)$.
- iii) The inverse element of $(a; b) \neq (0; 0)$ is $(a; b)^{-1} = \left(\frac{a}{a^2+b^2}; \frac{-b}{a^2+b^2}\right)$.



The canonical form of complex numbers

Let $i = (0; 1)$. Hence,

- i) $z = (a; b) = (a; 0) + (0; b) = a(1; 0) + b(0; 1) = a + bi$
- ii) $i^2 = (0; 1)(0; 1) = (-1; 0) = -1$

Definition

For every complex number $z \in \mathbb{C}$, $z = a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$. Now $z = a + bi$ is called the canonical form of the complex number z .

- i) The real number a is called the real part of z and denoted by $a = \operatorname{Re}(z)$.
- ii) The real number b is called the imaginary part of z and denoted by $b = \operatorname{Im}(z)$.
- iii) i is called the imaginary unit.



The canonical form of complex numbers

Operations of complex numbers in the canonical form

- i) Addition: $(a + bi) + (c + di) = (a + c) + (b + d)i$
- ii) Subtraction: $(a + bi) - (c + di) = (a - c) + (b - d)i$
- iii) Multiplication: $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$
- iv) Division: $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$, where $c^2 + d^2 \neq 0$



The conjugate and the modulus of a complex number

Let $z = a + bi$, where $a, b \in \mathbb{R}$, be a complex number

The conjugate of a complex number

The conjugate of z is defined by $\bar{z} = a - bi$

The modulus of a complex number

The modulus of z is defined by $|z| = \sqrt{a^2 + b^2}$

Remark

- | | |
|---|------------------------|
| 1) $ z = \bar{z} $ | 4) $\bar{\bar{z}} = z$ |
| 2) $z + \bar{z} = 2\operatorname{Re}(z) = 2a$ | 5) $z\bar{z} = z ^2$ |
| 3) $\frac{1}{z} = \frac{\bar{z}}{ z ^2}$ | |



Properties of a complex number

Properties

- i) $z_1 + z_2 = z_2 + z_1$; $z_1 z_2 = z_2 z_1$
- ii) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$; $z_1(z_2 z_3) = (z_1 z_2)z_3$
- iii) $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$
- iv) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$; $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$
- v) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}}$
- vi) $|z_1 z_2| = |z_1| |z_2|$; $|z_1 + z_2| \leq |z_1| + |z_2|$

Examples

E1) Compute $A = \frac{1+2i}{4-3i} + \frac{1}{2i} - \frac{3}{4}$

E2) Given $|z_1| = 1$ and $\forall z_2 \neq z_1$. Prove that $\left| \frac{z_1 - z_2}{1 - \overline{z_1} z_2} \right| = 1$



The polar (trigonometric) form of complex numbers

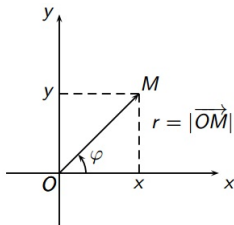
Each complex number $z = a + bi$ can be represented by a point $M(a, b)$ in the plane Oxy , where Ox is the real axis and Oy is the imaginary axis.

Let $\begin{cases} r = |\overrightarrow{OM}| \\ \varphi = (Ox, \overrightarrow{OM}) \end{cases}$. Now $z = a + bi = r(\cos \varphi + i \sin \varphi)$, where

i) $r = |z| = \sqrt{a^2 + b^2}$

ii) φ is called the Argument of z , denoted by $\text{Arg}(z)$. Moreover,

$$\cos \varphi = \frac{a}{\sqrt{a^2 + b^2}}; \sin \varphi = \frac{b}{\sqrt{a^2 + b^2}}$$



The polar (trigonometric) form of complex numbers

Examples

Determine the polar form of the following complex numbers

a) $z = \sqrt{3} + i$

b) $z = -\sqrt{3} - i$

c) $z = -2$

d) $z = 3i$



Operations of complex numbers in the polar form

Operations of complex numbers in the polar form

Let $z_1 = r_1(\cos \varphi_1 + i \sin \varphi_1)$, $z_2 = r_2(\cos \varphi_2 + i \sin \varphi_2)$ be two complex numbers.

- i) Multiplication: $z_1 z_2 = r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2))$
Hence $|z_1 z_2| = |z_1| |z_2|$, $\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$.
- ii) Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2))$
Hence $|\frac{z_1}{z_2}| = \frac{|z_1|}{|z_2|}$, $\text{Arg}(\frac{z_1}{z_2}) = \text{Arg}(z_1) - \text{Arg}(z_2)$.

Examples

Given $z_1 = 6(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12})$, $z_2 = 4(\cos \frac{-\pi}{6} + i \sin \frac{-\pi}{6})$. Compute $z_1 \cdot z_2$ and $\frac{z_1}{z_2}$



Operations of complex numbers in the polar form

Remarks

If $z = r(\cos \varphi + i \sin \varphi)$, then

① $\bar{z} = r(\cos(-\varphi) + i \sin(-\varphi))$

② $z^{-1} = \frac{1}{r}(\cos(-\varphi) + i \sin(-\varphi))$

Exponentiation (Moivre's formula)

$$z = r(\cos \varphi + i \sin \varphi) \Rightarrow z^n = r^n(\cos n\varphi + i \sin n\varphi)$$

Clearly, $|z^n| = |z|^n$

Example

E1) Compute $z = (\sqrt{3} + i)^{2019}$

Operations of complex numbers in the polar form

Definition

n -roots of a complex number z is denoted by $\sqrt[n]{z}$. If $z = r(\cos \varphi + i \sin \varphi) \neq 0$, then

$$\sqrt[n]{z} = \sqrt[n]{r} \left(\cos \frac{\varphi + k2\pi}{n} + i \sin \frac{\varphi + k2\pi}{n} \right), k = \overline{0, n-1}$$

Each non-zero complex number has exactly n different n -roots.

Examples

Compute

E1) $\sqrt[3]{8(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})}$

E2) $\sqrt[3]{8}$

E3) $\sqrt{1+i}$

The field of complex numbers

Examples

E1) Compute the following complex numbers

a) $(1 + i\sqrt{3})^{2019}$

b) $(2 + i\sqrt{12})^5(\sqrt{3} - i)^{11}$

E2) Solve the following equations

a) $z^4 - 3iz^2 + 4 = 0$

b) $z^6 - 7z^3 - 8 = 0$

c) $\frac{(z+i)^4}{(z-i)^4} = 1$

d) $z^8(\sqrt{3} + i) = 1 - i$



The end

Thanks for your attention!

